

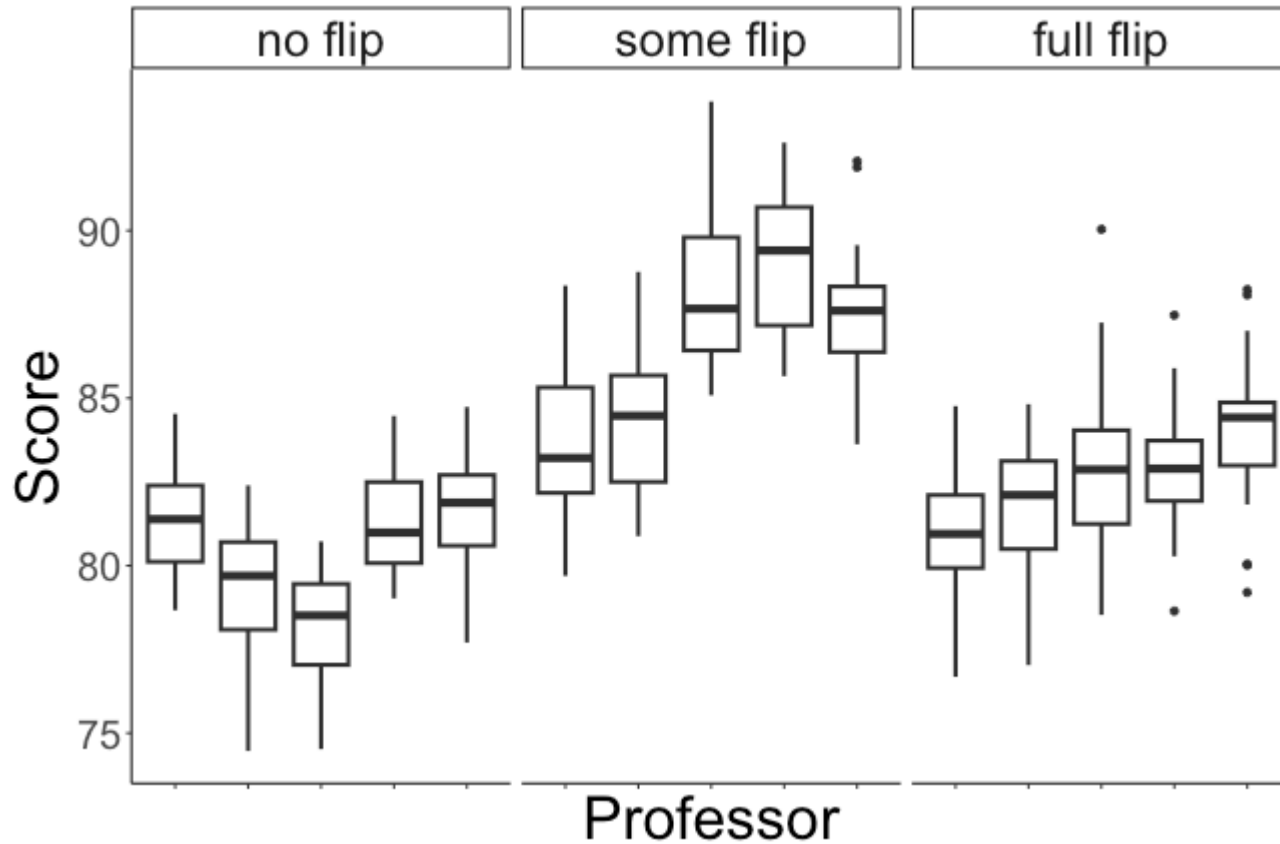
Beginning linear mixed effects models

Data: flipped classrooms?

Data set has 375 rows (one per student), and the following variables:

- + professor: which professor taught the class (1 -- 15)
- + style: which teaching style the professor used (no flip, some flip, fully flipped)
- + score: the student's score on the final exam

Visualizing the data



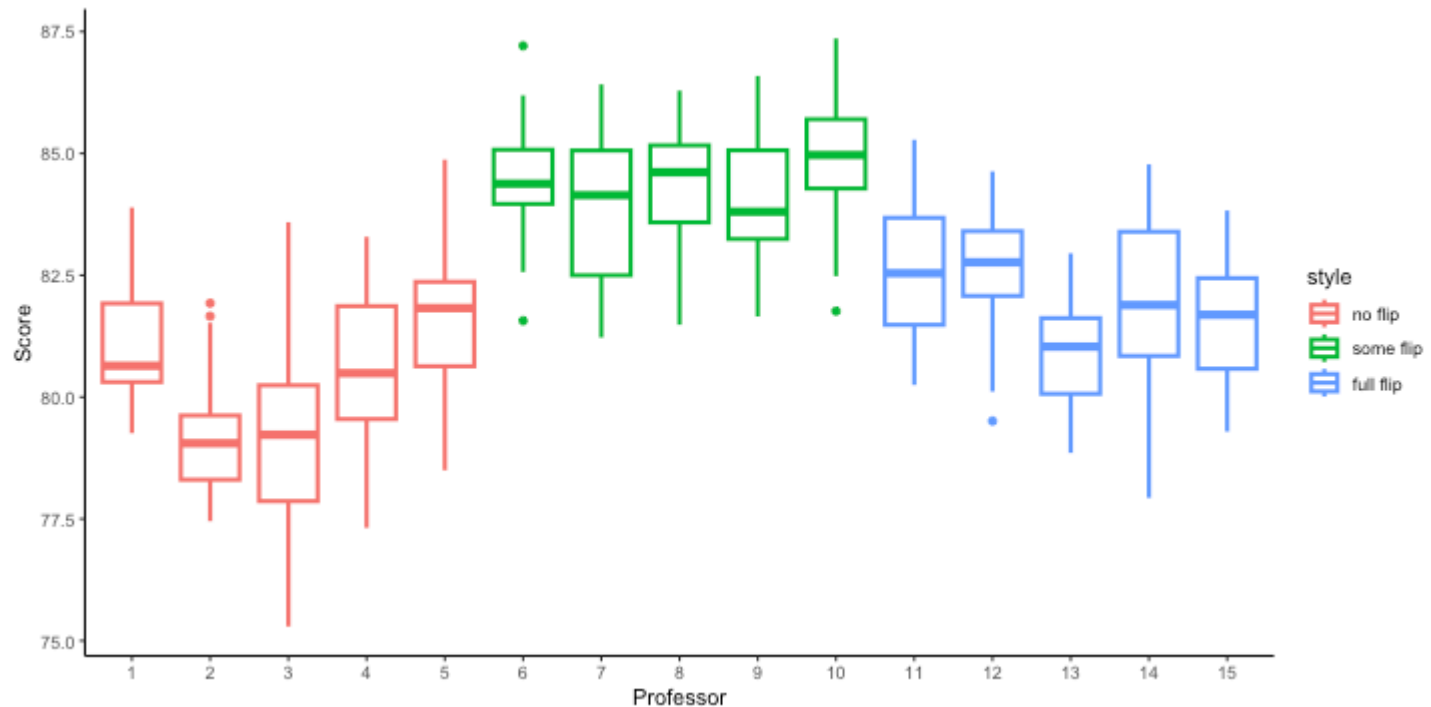
Mixed effects model

Linear mixed effects model: Let $Score_{ij}$ be the score of student j in class i

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

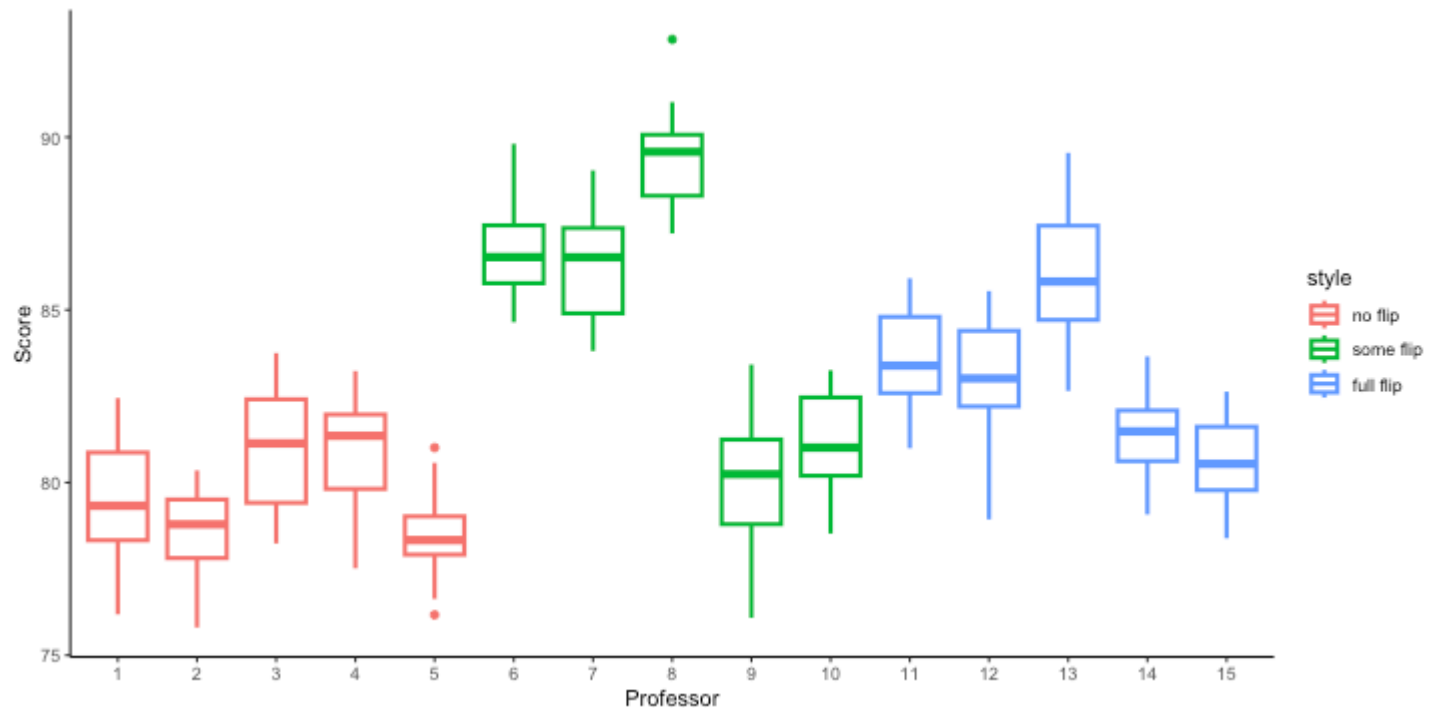
$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2) \quad u_i \stackrel{iid}{\sim} N(0, \sigma_u^2)$$

Last time: Class activity



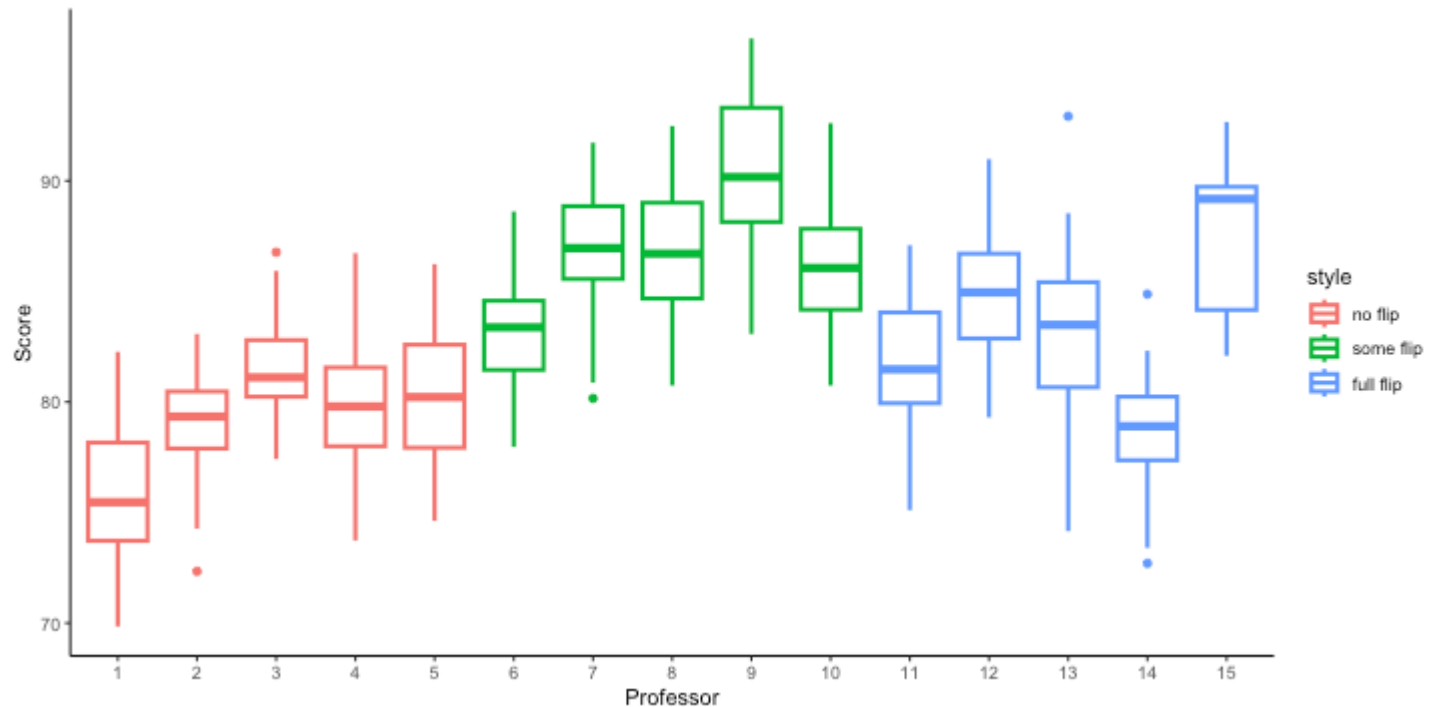
How does the plot change if I increase σ_u^2 ?

Class activity



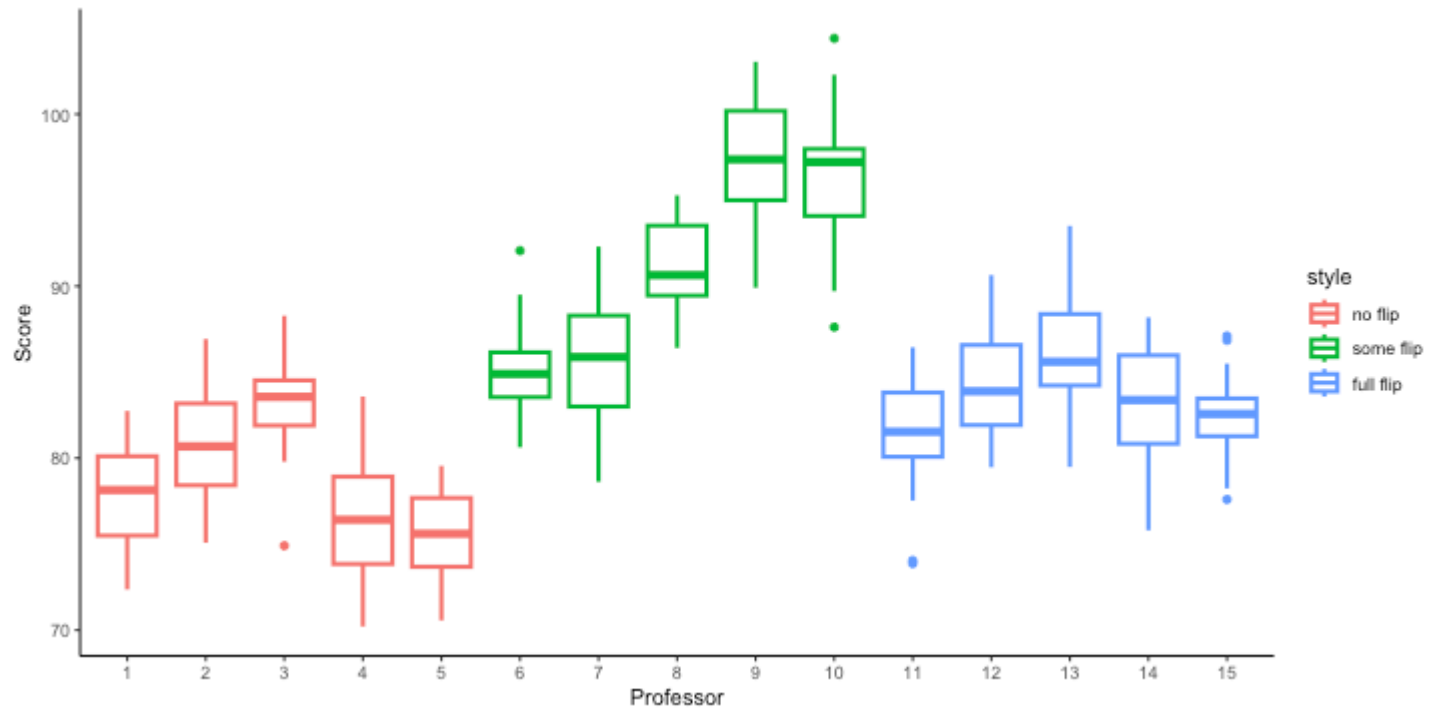
How does the plot change if I increase σ_{ϵ}^2 ?

Class activity



How does the plot change if I increase β_1 ?

Class activity



Class activity

Work on the class activity (handout), then we will discuss as a class.

Class activity

Why is a mixed effect model useful for this data?

Class activity

What is the population model?

Class activity

What is the population model?

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental j in neighborhood i

- + u_i is a *random intercept*
- + We use subscripts i and j for $Price_{ij}$, $Satisfaction_{ij}$, and ε_{ij} because they are different for every observation in the data
- + We only need a subscript i (neighborhood) for u_i , because there is one random intercept per neighborhood

Class activity

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental j in neighborhood i

What are the effect of interest, group effect, and individual effect?

Class activity

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

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where $Price_{ij}$ is the price of rental j in neighborhood i

What are the effect of interest, group effect, and individual effect?

- + effect of interest: β_1 (slope for relationship between satisfaction and price)
- + group effect: u_i (random effect for neighborhood)
- + individual effect: ε_{ij} (variation between rentals in a neighborhood)

Fitting mixed effects models

```
library(lme4)
m1 <- lmer(score ~ style + (1|professor),
           data = teaching)
summary(m1)
```

- + lme4 is the R package we will use to fit mixed effects models
- + lmer is like the lm function, but for mixed effects
- + style is the teaching style (fixed effects)
- + (1|professor) indicates we have a random intercept (the 1 indicates the intercept) for professor (indicated by |professor)

Fitting mixed effects models

```
library(lme4)
```

```
m1 <- lmer(score ~ style + (1|professor),  
           data = teaching)  
summary(m1)
```

Random effects:

Groups	Name	Variance	Std.Dev.
professor	(Intercept)	21.365	4.622
Residual		4.252	2.062

Number of obs: 375, groups: professor, 15

+ $\hat{\sigma}_{\varepsilon}^2 = 4.25$

+ $\hat{\sigma}_u^2 = 21.37$

Fitting mixed effects models

```
m1 <- lmer(score ~ style + (1|professor),  
            data = teaching)  
summary(m1)
```

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	77.657	2.075	37.419
stylesome flip	11.073	2.935	3.773
stylefull flip	2.805	2.935	0.956

+ $\hat{\beta}_0 = 77.66$

+ $\hat{\beta}_1 = 11.07$

+ $\hat{\beta}_2 = 2.81$

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2) \quad u_i \stackrel{iid}{\sim} N(0, \sigma_u^2)$$

$$+ \quad \hat{\beta}_0 = 77.66, \quad \hat{\beta}_1 = 11.07, \quad \hat{\beta}_2 = 2.81$$

$$+ \quad \hat{\sigma}_\varepsilon^2 = 4.25, \quad \hat{\sigma}_u^2 = 21.37$$

How do I interpret $\hat{\beta}_0 = 77.66$?

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

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$$+ \quad \hat{\beta}_0 = 77.66, \quad \hat{\beta}_1 = 11.07, \quad \hat{\beta}_2 = 2.81$$

$$+ \quad \hat{\sigma}_\varepsilon^2 = 4.25, \quad \hat{\sigma}_u^2 = 21.37$$

How do I interpret $\hat{\beta}_1 = 11.07$ and $\hat{\beta}_2 = 2.81$?

Interpretation

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$$+ \quad \hat{\beta}_0 = 77.66, \quad \hat{\beta}_1 = 11.07, \quad \hat{\beta}_2 = 2.81$$

$$+ \quad \hat{\sigma}_\varepsilon^2 = 4.25, \quad \hat{\sigma}_u^2 = 21.37$$

How do I interpret $\hat{\beta}_1 = 11.07$ and $\hat{\beta}_2 = 2.81$?

We expect that, on average, scores in some-flipped classes are 11.07 points higher than for no-flipped classes.

We expect that, on average, scores in fully-flipped classes are 2.81 points higher than for no-flipped classes.

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2) \quad u_i \stackrel{iid}{\sim} N(0, \sigma_u^2)$$

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How do I interpret $\hat{\sigma}_u^2 = 21.37$?

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$$+ \quad \hat{\sigma}_\varepsilon^2 = 4.25, \quad \hat{\sigma}_u^2 = 21.37$$

How do I interpret $\hat{\sigma}_u^2 = 21.37$?

The average score varies from professor to professor by about $4.62 (= \sqrt{21.37})$ points

Interpretation

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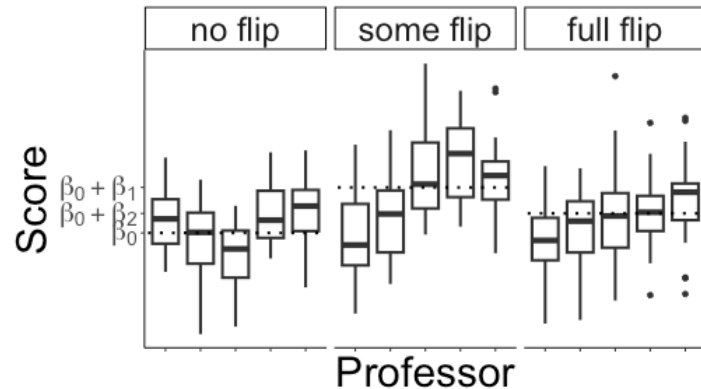
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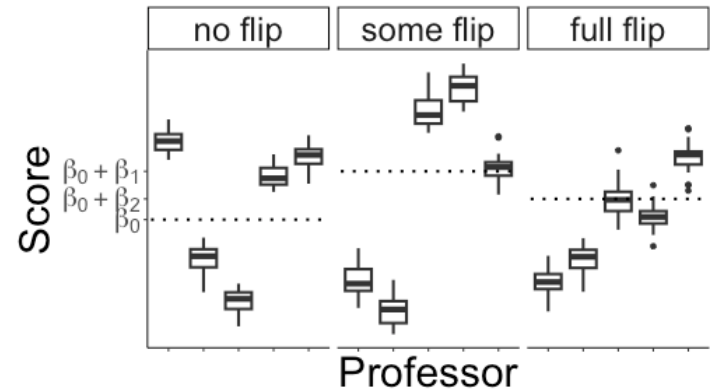
How do I interpret $\hat{\sigma}_\varepsilon^2 = 4.25$?

Within a class, students' scores vary by about 2.06 ($= \sqrt{4.25}$) points

Intra-class correlation



σ_ε^2 is large relative to σ_u^2



σ_ε^2 is small relative to σ_u^2

✚ Observations within a group are *more correlated* when σ_ε^2 is small relative to σ_u^2

✚ **Intra-class correlation:**

$$\rho_{group} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} = \frac{\text{between group variance}}{\text{total variance}}$$

Intra-class correlation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2) \quad u_i \stackrel{iid}{\sim} N(0, \sigma_u^2)$$

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$$+ \quad \hat{\sigma}_\varepsilon^2 = 4.25, \quad \hat{\sigma}_u^2 = 21.37$$

$$\hat{\rho}_{group} = \frac{21.37}{21.37 + 4.25} = 0.83$$

So 83% of the variation in student's scores can be explained by differences in average scores from class to class (after accounting for teaching style). That's huge!

Class activity

Work on the class activity (handout).

Class activity

Interpret the estimate fixed effect coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$.

Class activity

Interpret the estimate fixed effect coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$.

On average (across neighborhoods), we expect that the price of rental with 0 overall satisfaction is \$27.28.

For a fixed neighborhood, an increase of 1 point in overall satisfaction is associated with an increase of \$14.81 in rental price.

Class activity

Calculate and interpret the estimated intra-class correlation.

Class activity

Calculate and interpret the estimated intra-class correlation.

$$\hat{\rho}_{group} = \frac{1048}{1048 + 6762} = 0.134$$

About 13% of the variability in price can be explained by differences in the average price between neighborhoods (after accounting for overall satisfaction).